Build the donor Hamiltonian

Read from top to bottom: this is the actual execution order.

1 Validate inputs

Set Nx, Ny, Nz; choose whether to include spin.

```
shape = broadcast_to(...); shape.astype(int)
```

2 Build the pristine Si box

Get the sparse Hamiltonian and all atom positions.

```
h, positions = make_real_space(..., periodic=True)
```

3 Select and validate the Bi site

Use the supplied index, or choose the atom nearest the centre.

```
donor_index = argmin(distance_to_centre_squared)
```

4 Measure periodic donor distances

Wrap each displacement, then calculate its length.

```
d -= lengths*rint(d/lengths); radius = norm(d, axis=1)
```

5 Shift the surrounding Si sites

Evaluate the tail only away from Bi; shift every local state.

```
potential[away] = donor_tail(...); h.setdiag(...)
```

6 Construct the local Bi correction

Replace Si orbital energies and SOC at the donor site.

```
correction = bi_onsite(...) - onsite(...)
```

7 Insert the sparse correction

Map local indices to global indices and add the correction.

```
offset = donor_index*norb; h += sparse_correction
```

8 Clean up and return

Return H, positions, donor index, radii and scalar potential.

```
h.eliminate_zeros(); return h, positions, ...
```

Assembly ends here. solve_near() calculates eigenstates later.

What changes physically?

Three visual reminders for the geometry, potential and matrix assembly.

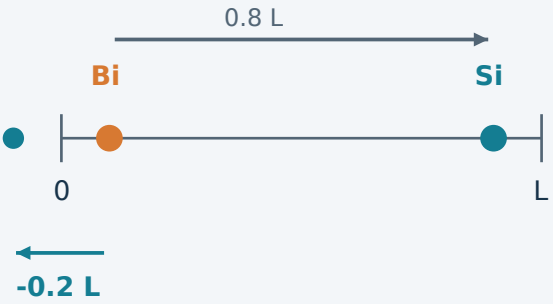
1 One Si atom becomes Bi



A substitutional impurity adds no atom.
The chosen site keeps its position,
but receives a new onsite block.

Blue: Si Orange: Bi
Diamond-lattice fragment shown;
periodic wrap bonds are omitted.

2 Use the nearest periodic image



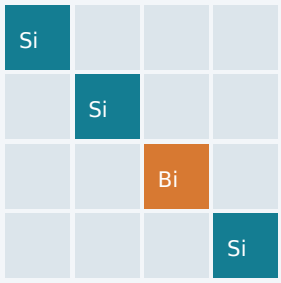
Example along one axis:
d = 0.8 L becomes d = -0.2 L.

Apply wrapping on x, y and z.
Use the resulting radius in donor_tail().

This is a minimum-image potential,
not a sum over all donor images.

The extra blue dot is an equivalent periodic image of the Si site.

3 Two onsite changes; same hopping



Atom-block schematic

Surrounding atoms:

H_ii <- H_Si + V(r_i) I

Bi site:

H_DD <- H_Bi

Off-diagonal atom blocks:

H_ij unchanged for i != j

Reminder: check box-size convergence before interpreting isolated-donor energies.